2D Arrays

LEVEL 1: Matrix Basics

Program 1: Input and Display Matrix Write a program that inputs an M×N matrix and displays it in proper matrix format.

Program 2: Display Matrix Row by Row Write a program that displays each row of matrix on a separate line with row number.

Program 3: Display Matrix Column by Column Write a program that displays each column of matrix separately with column number.

Program 4: Sum of All Elements Write a program that calculates sum of all elements in the matrix.

Program 5: Count Even and Odd Elements Write a program that counts how many even and odd numbers are in the matrix.

Program 6: Count Positive, Negative, and Zero Write a program that counts positive, negative, and zero elements separately.

Program 7: Find Maximum Element in Matrix Write a program that finds the largest element in entire matrix.

Program 8: Find Minimum Element in Matrix Write a program that finds the smallest element in entire matrix.

Program 9: Search Element in Matrix Write a program that searches for element X and displays its position (row, column). Display "Not Found" if absent.

Program 10: Count Occurrences of Element Write a program that counts how many times element X appears in the matrix.

Program 11: Sum of Each Row Write a program that calculates and displays sum of each row separately.

Program 12: Sum of Each Column Write a program that calculates and displays sum of each column separately.

Program 13: Average of Each Row Write a program that calculates average of elements in each row.

Program 14: Average of Each Column Write a program that calculates average of elements in each column.

Program 15: Overall Average of Matrix Write a program that calculates average of all elements in matrix.

LEVEL 2: Row & Column Operations

Program 16: Find Row with Maximum Sum Write a program that finds which row has maximum sum of elements.

Program 17: Find Row with Minimum Sum Write a program that finds which row has minimum sum of elements.

Program 18: Find Column with Maximum Sum Write a program that finds which column has maximum sum.

Program 19: Find Column with Minimum Sum Write a program that finds which column has minimum sum.

Program 20: Maximum Element in Each Row Write a program that finds and displays maximum element in each row.

Program 21: Minimum Element in Each Row Write a program that finds and displays minimum element in each row.

Program 22: Maximum Element in Each Column Write a program that finds and displays maximum element in each column.

Program 23: Minimum Element in Each Column Write a program that finds and displays minimum element in each column.

Program 24: Swap Two Rows Write a program that swaps row i with row j (indices given by user).

Program 25: Swap Two Columns Write a program that swaps column i with column j.

Program 26: Reverse Each Row Write a program that reverses elements in each row individually.

Program 27: Reverse Each Column Write a program that reverses elements in each column individually.

LEVEL 3: Diagonal & Boundary Operations

Program 28: Print Main Diagonal Elements Write a program that prints all elements on main diagonal (top-left to bottom-right).

Program 29: Print Secondary Diagonal Elements Write a program that prints all elements on secondary diagonal (top-right to bottom-left).

Program 30: Sum of Main Diagonal Write a program that calculates sum of main diagonal elements.

Program 31: Sum of Secondary Diagonal Write a program that calculates sum of secondary diagonal elements.

Program 32: Sum of Both Diagonals Write a program that calculates sum of both diagonals together (handle center element in odd-sized matrix).

Program 33: Print Diagonal Difference Write a program that finds absolute difference between sums of two diagonals.

Program 34: Print Boundary Elements Write a program that prints all boundary (border) elements of matrix.

Program 35: Sum of Boundary Elements Write a program that calculates sum of all boundary elements.

Program 36: Print Upper Triangle Elements Write a program that prints all elements above main diagonal.

Program 37: Print Lower Triangle Elements Write a program that prints all elements below main diagonal.

Program 38: Sum of Upper Triangle Write a program that calculates sum of all elements above main diagonal.

Program 39: Sum of Lower Triangle Write a program that calculates sum of all elements below main diagonal.

LEVEL 4: Matrix Transformations

Program 40: Transpose of Matrix Write a program that finds transpose of matrix (swap rows and columns). Store in new matrix.

Program 41: Transpose Square Matrix In-Place Write a program that transposes square matrix by modifying original matrix.

Program 42: Rotate Matrix 90° Clockwise Write a program that rotates matrix 90 degrees clockwise. Example: [[1,2],[3,4]] → [[3,1],[4,2]].

Program 43: Rotate Matrix 90° Anti-Clockwise Write a program that rotates matrix 90 degrees counter-clockwise.

Program 44: Rotate Matrix 180° Write a program that rotates matrix 180 degrees.

Program 45: Flip Matrix Horizontally Write a program that flips matrix horizontally (reverse each row).

Program 46: Flip Matrix Vertically Write a program that flips matrix vertically (reverse row order).

Program 47: Mirror Matrix Along Main Diagonal Write a program that mirrors matrix along main diagonal (transpose).

Program 48: Mirror Matrix Along Secondary Diagonal Write a program that mirrors along secondary diagonal.

Program 49: Reverse Entire Matrix Write a program that reverses all elements treating matrix as 1D array.

LEVEL 5: Matrix Arithmetic

Program 50: Add Two Matrices Write a program that adds two matrices of same size element-wise.

Program 51: Subtract Two Matrices Write a program that subtracts matrix B from matrix A.

Program 52: Multiply Matrix by Scalar Write a program that multiplies every element of matrix by a scalar value.

Program 53: Multiply Two Matrices Write a program that multiplies two compatible matrices (rows of A = columns of B).

Program 54: Check if Two Matrices are Equal Write a program that checks if two matrices have same dimensions and same elements.

Program 55: Element-wise Multiplication (Hadamard Product) Write a program that multiplies corresponding elements of two matrices of same size.

Program 56: Power of Matrix Write a program that calculates matrix raised to power N (repeated multiplication).

Program 57: Check if Matrix is Scalar Matrix Write a program that checks if matrix is scalar (diagonal elements equal, others zero).